KIRUBANANDHAN V

Python Backend Developer | Fresher

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GitHub | LinkedIn

Professional Summary

Entry-level Python Backend Developer with hands-on project experience building REST APIs using FastAPI, SQLAlchemy, Pydantic, and SQLite. Developed an inventory and order management API with JWT authentication, database migrations, transaction handling, and automated testing using pytest. Familiar with relational database design and SQL querying, including joins, CTEs, aggregate functions, and window functions. Seeking an opportunity to contribute to backend development while continuing to strengthen software engineering skills.

Technical Skills

Programming Languages: Python, SQL ,C

Backend Development: FastAPI, REST API Development, JWT Authentication, Pydantic Validation, Swagger/OpenAPI

Databases and ORM: SQLite, SQLAlchemy 2.0, Alembic Migrations, Relational Schema Design, Primary and Foreign Keys, Constraints, Indexes

SQL: JOINs, LEFT JOIN, GROUP BY, Aggregate Functions, Common Table Expressions (CTEs), Window Functions, Subqueries, CASE, COALESCE

Backend Concepts: Exception Handling, HTTP Error Responses, Database Transactions, Commit/Rollback

Testing and Tools: pytest, Git, GitHub, VS Code

Projects

InventoryX – Inventory and Order API GitHub Repository

Python, FastAPI, SQLAlchemy, Alembic, Pydantic, JWT, SQLite, pytest

- Built a FastAPI inventory and order REST API with 20 operations across 13 paths and interactive Swagger/OpenAPI documentation.
- Designed SQLAlchemy 2.0 models for six related tables and managed database schema changes with Alembic migrations.
- Implemented JWT authentication using HS256 and password hashing with salted PBKDF2-SHA256.
- Added Pydantic request/response validation and CRUD endpoints for products and categories.
- Implemented stock adjustments, order placement and cancellation, inventory validation, stock history, and rollback handling for database integrity errors.
- Wrote 10 pytest tests to verify API behavior.

E-commerce Database Management System GitHub Repository

SQL, SQLite

- Designed a four-table relational schema for customers, products, orders, and order items using primary keys, foreign keys, composite keys, and CHECK constraints.
- Created six indexes and wrote 16 SQL queries for inventory, customer-spending, and sales analysis.
- Applied JOINs, LEFT JOIN, GROUP BY, aggregate functions, CTEs, RANK() window functions, subqueries, CASE, and COALESCE.

Student Management REST API GitHub Repository

Python, FastAPI

- Developed a basic REST API with five GET endpoints using in-memory sample data.
- Implemented path parameters, query filters, summary calculations, and HTTP 404 error handling.

Education

Bachelor of Engineering (B.E.) – Electronics and Communication Engineering

IFET College of Engineering, Villupuram 2022–2026

Anna University CGPA: 7.14/10